

Hands-On-Learning: Building a Compliance & Cost Dashboard for Cloud Governance

Scenario

You are a cloud governance engineer asked to brief both the **CISO** and the **CFO** on your organization's AWS security posture. They want to see not only that controls (like encryption, tagging, and public-bucket prevention) are in place, but also how these measures affect **cost efficiency**.

Your task is to design a **compliance & cost dashboard** that merges technical guardrails with financial metrics. This dashboard will become a reusable artifact for audits, budget reviews, and executive updates.

Instructions for Learners

1 Preparation

Tools you may use:

- AWS Console (Cost Explorer, Config, Security Hub, QuickSight)

- Spreadsheet or visualization tool (Excel, Google Sheets, Tableau, Lucidchart)

Review these AWS features:

- Tags** (Owner, CostCenter, DataClass)

- AWS Config conformance packs**

- Cost Explorer with cost allocation tags**

- QuickSight dashboards**

2 Select Controls to Track

Choose at least **3 core security controls** to represent in your dashboard:

- MFA on root/admin accounts**

- Block Public S3 Buckets
- RDS Encryption with KMS
- Required Tags for cost allocation

For each control, note:

- Framework Alignment:** CIS / ISO / NIST mapping
- Business Value:** e.g., “Encryption → reduces regulatory risk”

3 Define Dashboard Metrics

Design metrics that speak to **both security and finance**:

- % of S3 buckets compliant with Block Public Access

- % of EC2/RDS resources encrypted at rest

- % of resources with Owner/CostCenter tags applied

- Monthly savings from right-sizing instances or using lifecycle policies

Sketch how these metrics will be **displayed**: pie charts, compliance scorecards, cost trend graphs.

4 Build the Dashboard Prototype

If using AWS:

- In **Config**, enable rules like s3-bucket-public-read-prohibited.
- In **Cost Explorer**, group costs by CostCenter tag.
- Combine these into a **QuickSight dashboard**.

If using offline tools:

- Create a mockup in Excel or PowerPoint showing compliance % + cost trends.

Artifact: a dashboard prototype (screenshot, mockup, or slide).

5 Peer Review

Share your dashboard mockup with a peer.

Ask:

Does it clearly show **both compliance and cost outcomes**?

Would the CFO understand the financial impact?

Would the CISO trust the compliance evidence?

Revise based on feedback.

Reflection

- Which controls were easiest to link to cost outcomes?
- Which metrics were harder to quantify?
- How can you ensure the dashboard stays **current** as resources change monthly?

Final Outcome

By completing this lab, you have:

Designed a **compliance & cost dashboard** tailored to executives.

Linked cloud controls to **audit evidence** and **financial impact**.

Produced a peer-reviewed artifact suitable for boardroom presentations.

Next Steps

Automate report generation with AWS Config + Cost Explorer + QuickSight.

Expand the dashboard to cover **multi-cloud** or hybrid workloads.

Use the dashboard as input to a **monthly self-audit process**.

Reflection

Which responsibilities surprised you most (things you thought the provider handled but are actually customer tasks)?

How would this matrix change for a multi-cloud environment?

How could this matrix reduce risk in real-world onboarding and training?

Final Outcome

Created a **Shared Responsibility Matrix** for IaaS, PaaS, and SaaS.

Produced a peer-reviewed artifact suitable for training and executive communication.

Built a reusable tool to guide future cloud migrations and compliance efforts.

Next Steps

Expand the matrix to include **specific services** (EC2, RDS, S3, Google Workspace, etc.).

Layer in **framework mapping** (e.g., CIS, ISO, NIST) to show compliance overlap.

Use this matrix as a foundation for a **Cloud Security Onboarding Guide** in your organization.

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project



Use word processing software to create your report.

Ensure all sections of the project document structure are completed.

Proofread and edit your report for clarity and accuracy.



Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.



Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.